

A Parity Obstruction for Odd Approximate Hadamard Matrices

Working note – corrected version

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Abstract

For an $n \times n$ sign matrix A , let

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)},$$

with the value $+\infty$ when A is singular. We prove that every odd-order sign matrix satisfies

$$\kappa(A) \geq \sqrt{1 + \frac{n-1}{n^2}} + \frac{\sqrt{n-1}}{n}.$$

In particular, for every odd $n \geq 5$, one has $\kappa(A) > 1 + 1/\sqrt{n}$. This gives a negative answer to the concrete rate proposed in Problem 2 of Steinerberger’s January 2025 problem list. The proof uses only parity of row inner products and an elementary spectral variance inequality.

1 The spectral inequality

The following lemma isolates the linear-algebraic part of the argument.

Lemma 1 (Variance forces conditioning). *Let $G \in \mathbb{R}^{n \times n}$ be positive definite, with eigenvalues $\lambda_1, \dots, \lambda_n$. Put*

$$\mu = \frac{1}{n} \sum_{i=1}^n \lambda_i, \quad V = \frac{1}{n} \sum_{i=1}^n (\lambda_i - \mu)^2.$$

Then

$$\sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} \geq \sqrt{1 + \frac{V}{\mu^2}} + \frac{\sqrt{V}}{\mu}. \quad (1)$$

Proof. Write $m = \lambda_{\min}$, $M = \lambda_{\max}$, and $R = M/m$. For every $\lambda \in [m, M]$,

$$(M - \lambda)(\lambda - m) \geq 0.$$

Averaging over the eigenvalues gives the Bhatia–Davis variance bound

$$V \leq (M - \mu)(\mu - m). \quad (2)$$

For fixed R , substitute $M = Rm$ and maximize the right-hand side of (2) over $m \in [\mu/R, \mu]$. The maximum occurs at

$$m = \frac{\mu(R+1)}{2R},$$

and hence

$$V \leq \mu^2 \frac{(R-1)^2}{4R}.$$

If $K = \sqrt{R}$, then

$$\frac{K - K^{-1}}{2} \geq \frac{\sqrt{V}}{\mu}.$$

Setting $\delta = \sqrt{V}/\mu$ and solving $K^2 - 2\delta K - 1 \geq 0$ for $K \geq 1$ yields

$$K \geq \delta + \sqrt{1 + \delta^2},$$

which is exactly (1). □

2 The parity obstruction

Let

$$\kappa_{\pm}(n) = \min_{A \in \{-1,1\}^{n \times n}} \kappa(A).$$

Theorem 1. *For every odd positive integer n ,*

$$\kappa_{\pm}(n) \geq \sqrt{1 + \frac{n-1}{n^2}} + \frac{\sqrt{n-1}}{n}. \quad (3)$$

Proof. Fix $A \in \{-1,1\}^{n \times n}$. There is nothing to prove if A is singular, so suppose it is invertible and set $G = AA^{\top}$. Every diagonal entry of G equals n . Since n is odd, the inner product of any two distinct rows of A is an odd integer; in particular,

$$|G_{ij}| \geq 1 \quad (i \neq j). \quad (4)$$

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of G , then

$$\frac{1}{n} \sum_i \lambda_i = \frac{\text{tr } G}{n} = n.$$

Moreover, using (4),

$$\sum_{i=1}^n (\lambda_i - n)^2 = \text{tr}(G^2) - n^3 = \sum_{i \neq j} G_{ij}^2 \geq n(n-1).$$

Thus the eigenvalue variance in Lemma 1 satisfies $V \geq n-1$, while $\mu = n$. Since

$$\kappa(A) = \sqrt{\frac{\lambda_{\max}(G)}{\lambda_{\min}(G)}},$$

Lemma 1 gives (3). □

Corollary 1. *For every odd $n \geq 5$ and every $A \in \{-1,1\}^{n \times n}$,*

$$\kappa(A) > 1 + \frac{1}{\sqrt{n}}. \quad (5)$$

Consequently, the proposed bound in Steinerberger's Problem 2 cannot hold for every n .

Proof. Put $t = n^{-1/2}$ and

$$\delta = \frac{\sqrt{n-1}}{n} = t\sqrt{1-t^2}.$$

The right-hand side of (3) is the unique $K \geq 1$ satisfying

$$\frac{K - K^{-1}}{2} = \delta.$$

Since $K \mapsto (K - K^{-1})/2$ is strictly increasing on $[1, \infty)$, it is enough to check

$$t\sqrt{1-t^2} > \frac{(1+t) - (1+t)^{-1}}{2} = \frac{t(2+t)}{2(1+t)}.$$

After cancelling $t > 0$ and squaring, this is equivalent to

$$t(4 - t - 8t^2 - 4t^3) > 0.$$

For $n \geq 5$, one has $0 < t \leq 1/\sqrt{5}$, and

$$4 - t - 8t^2 - 4t^3 \geq \frac{12}{5} - \frac{9}{5\sqrt{5}} > 0.$$

This proves (5). □

The lower bound has the expansion

$$\kappa_{\pm}(n) \geq 1 + \frac{1}{\sqrt{n}} + \frac{1}{2n} + O(n^{-3/2}) \quad (n \rightarrow \infty \text{ through odd integers}). \quad (6)$$

3 Relation to recent work and novelty status

Steinerberger asked whether, for every n , there is a sign matrix with condition number at most $1 + 1/\sqrt{n}$ [1]. After that problem-list version, Alexeev, Jasper, and Mixon proved the qualitative statement $\kappa_{\pm}(n) \rightarrow 1$ and constructed explicit infinite families [2]. They also proved, for all sufficiently large $n \not\equiv 0 \pmod{4}$, the lower bound

$$\kappa_{\pm}(n) \geq 1 + c \frac{\log n}{n}$$

by a Ramsey-theoretic argument. For odd n , (3) strengthens that rate to $1 + \Omega(n^{-1/2})$ and rules out the particular constant proposed in the 2025 list.

A targeted search found no occurrence of (3) in the approximate-Hadamard literature, and it does not appear in the November 2025 paper, whose lower bound and list of follow-up problems would both make it directly relevant. This is a preliminary novelty check rather than independent confirmation; the authors of that paper and Steinerberger should be contacted before the observation is advertised as new.

References

- [1] S. Steinerberger, *Some Open Problems*, Problem 2, version dated 6 January 2025.
- [2] B. Alexeev, J. Jasper, and D. G. Mixon, *Asymptotically optimal approximate Hadamard matrices*, arXiv:2511.14653 (2025).
- [3] X. Dong and M. Rudelson, *Approximately Hadamard matrices and Riesz bases in random frames*, Int. Math. Res. Not. IMRN (2024), 2044–2065.
- [4] S. Steinerberger, *A note on approximate Hadamard matrices*, Des. Codes Cryptogr. 92 (2024), 3125–3131; arXiv:2402.13202.